

2020-2021 春 微积分 (I) -2 期末考试 A 卷参考答案

一 填空题 (每小题 4 分, 共 20 分)

1. 曲线 $x = e^t, y = t^2 + \sin t, z = 2t + 1$ 上过点 $(1, 0, 1)$ 的法平面方程为 $x + y + 2z = 3$

解: 该点对应 $t = 0$, 切向量为 $(1, 1, 2)$, 法平面 $x + y + 2z - 3 = 0$.

2. 设 $z = 9 + (x+1)^y$, 则 $\left(\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\right)\bigg|_{\substack{x=1 \\ y=1}} = (1 + 2\ln 2)$.

解: $\left(\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\right)\bigg|_{\substack{x=1 \\ y=1}} = \left[y(x+1)^{y-1}(x+1)' + (x+1)^y \ln(x+1)\right]\bigg|_{\substack{x=1 \\ y=1}} = 1 + 2\ln 2$.

3. 曲线积分 $\oint_{x^2+y^2=1} (1+x^2) ds = (3\pi)$.

解: $\oint_{x^2+y^2=1} (1+x^2) ds = \oint_{x^2+y^2=1} \left(1 + \frac{x^2+y^2}{2}\right) ds = 3\pi$.

4. 微分方程 $(1+y)^2 \frac{dy}{dx} + x^3 = 0$ 满足 $y(1) = 0$ 的特解为 $(4(1+y)^3 + 3x^4 = 7)$.

解: $(1+y)^2 dy + x^3 dx = 0, 4(1+y)^3 + 3x^4 = 7 (C=7)$.

5. 把周期为 2π 的周期函数 $f(x) = \begin{cases} \sin x - x, & -\pi \leq x < 0 \\ x - \cos x, & 0 \leq x < \pi \end{cases}$ 展开成傅里叶数,

设 $s(x)$ 是该傅里叶级数的和函数, 则 $s(\pi) = (\pi - \frac{1}{2})$.

解: $s(\pi) = \frac{1}{2}[f(-\pi+0) + f(\pi-0)] = \pi + \frac{1}{2}$.

二. 计算题 (每小题 7 分, 共 28 分)

6. 设二元函数 $z = z(x, y)$ 由方程 $xz + xy - x^2 + ye^z = 1$ 所确定,

求 $\frac{\partial z}{\partial y}\bigg|_{(1,1)}, \frac{\partial^2 z}{\partial y^2}\bigg|_{(1,1)}$.

解: 当 $x = 1, y = 1$, 由 $z + e^z - 1 = 0$, 得 $z(1, 1) = 0$;

求导: $x + e^z + (x + ye^z) \frac{\partial z}{\partial y} = 0$, 得 $\frac{\partial z}{\partial y}(1, 1) = -1$.

再求导: $2e^z \frac{\partial z}{\partial y} + ye^z \left(\frac{\partial z}{\partial y}\right)^2 + (x + ye^z) \frac{\partial^2 z}{\partial y^2} = 0$, 得 $\frac{\partial^2 z}{\partial y^2}(1, 1) = \frac{1}{2}$.

7. 计算 $\iint_{x^2+y^2 \leq 4} (2+ye^{|y|}+\sqrt{4-x^2})dxdy$.

解: 原式 $= 8\pi + 0 + \int_{-2}^2 \sqrt{4-x^2} dx \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dy = 8\pi + \frac{64}{3}$.

8. 计算曲线积分 $\int_L (1+3x^2+e^x \sin y)dx + (2y+e^x \cos y)dy$, 其中 L 是有向曲线:

$x = 2t - t^2, y = 2 \arctan t$ 从 $t = 0$ 到 $t = 1$ 的曲线段.

解: $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, 积分与路径无关;

$$\int_L (1+3x^2+e^x \sin y)dx + (2y+e^x \cos y)dy$$

$$= \int_{(0,0)}^{(1,\frac{\pi}{2})} d(x+x^3+e^x \sin y+y^2) = 2+e+\frac{\pi^2}{4}.$$

9. 求初值问题 $\begin{cases} y'' - y' + 2y = (x-1)e^x \\ y(0) = y'(0) = 1 \end{cases}$ 的特解.

解: (1) 特征方程为: $r^2 - r + 2 = 0$, 特征根 $r_1 = \frac{1+j\sqrt{7}}{2}, r_2 = \frac{1-j\sqrt{7}}{2}$,

对应齐次方程的通解为 $Y = e^{\frac{x}{2}} (C_1 \cos \frac{\sqrt{7}}{2} x + C_2 \sin \frac{\sqrt{7}}{2} x)$.

(2) $\lambda = 1$ 不是特征根, 设非齐次方程的特解 $y^* = (Ax+B)e^x$,

代入得 $A = \frac{1}{2}, B = -\frac{3}{4}$, 原方程的通解为

$$Y = e^{\frac{x}{2}} (C_1 \cos \frac{\sqrt{7}}{2} x + C_2 \sin \frac{\sqrt{7}}{2} x) + e^x (\frac{1}{2} x - \frac{3}{4}).$$

(3) 把 $y(0) = y'(0) = 1$ 代入通解, 得 $C_1 = \frac{7}{4}, C_2 = \frac{3\sqrt{7}}{28}$,

满足初始条件的特解为 $y = e^{\frac{x}{2}} (\frac{7}{4} \cos \frac{\sqrt{7}}{2} x + \frac{3\sqrt{7}}{28} \sin \frac{\sqrt{7}}{2} x) + e^x (\frac{1}{2} x - \frac{3}{4})$.

三. 解答题 (每小题 9 分, 共 18 分)

10. 设二元函数 $f(x,y) = \begin{cases} \frac{xy^2}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0) \end{cases}$

(1) 求 $\frac{\partial f}{\partial x}(0,0)$ 和 $\frac{\partial f}{\partial y}(0,0)$; (2) $f(x,y)$ 在原点 $(0,0)$ 是否可微?

(3) 求 $f(x, y)$ 在原点 $(0, 0)$ 沿方向 $l(\frac{4}{5}, \frac{3}{5})$ 的方向导数 $\frac{\partial f}{\partial l}(0, 0)$.

解: (1) $\frac{\partial f}{\partial x}(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - 0}{\Delta x} = 0, \frac{\partial f}{\partial y}(0, 0) = 0.$

(2) $\because \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{[f(\Delta x, \Delta y) - 0] - (0\Delta x + 0\Delta y)}{\sqrt{\Delta x^2 + \Delta y^2}} = \lim_{\rho \rightarrow 0} \frac{\rho^3 \cos \theta \sin^2 \theta}{\rho^3}$

极限不存在, 因此, 在原点不可微.

(3) $\frac{\partial f}{\partial l}(0, 0) = \lim_{\rho \rightarrow 0} \frac{f(\frac{4}{5}\rho, \frac{3}{5}\rho) - 0}{\rho} = \lim_{\rho \rightarrow 0} \frac{\rho^3 \frac{4}{5}(\frac{3}{5})^2}{\rho^3} = \frac{36}{125}.$

11. 设 Σ 由锥面 $z = \sqrt{x^2 + y^2}$ 和上半球面 $z = \sqrt{4 - x^2 - y^2}$ 组成, 方向为外侧.

计算曲面积分 $\iint_{\Sigma} \cos(y^2 + z) dy dz + (3x^2 y + y^3) dz dx + (xy + z^3) dx dy.$

解: 由高斯公式,

$$\begin{aligned} \text{原式} &= \iiint_{\Omega} 3(x^2 + y^2 + z^2) dx dy dz \\ &= 3 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{4}} d\varphi \int_0^2 r^2 r^2 \sin \varphi dr = \frac{96\pi}{5} (2 - \sqrt{2}). \end{aligned}$$

四 应用题 (每小题 8 分, 共 24 分)

12. 空间物体 Ω 由两曲面 $z = \sqrt{2 - x^2 - y^2}$ 和 $x^2 + y^2 = z$ 围成, 求它的体积.

解: $V = \iint_{x^2 + y^2 \leq 1} (\sqrt{2 - x^2 - y^2} - x^2 - y^2) dx dy$

$$= \int_0^{2\pi} d\theta \int_0^1 (\sqrt{2 - r^2} - r^2) r dr = \frac{1}{6} (8\sqrt{2} - 7)\pi.$$

13. 把函数 $f(x) = \frac{x}{1-x} + \ln(1-x)$ 在 $x=0$ 处展开成泰勒级数, 并求 $f^{(2021)}(0)$.

解: $f(x) = \frac{x}{1-x} - \int_0^x \frac{1}{1-x} dx = \sum_{n=1}^{\infty} x^n - \int_0^x \sum_{n=0}^{\infty} x^n dx$

$$= \sum_{n=1}^{\infty} (1 - \frac{1}{n}) x^n, \quad x \in (-1, 1).$$

$\therefore a_{2021} = 1 - \frac{1}{2021} = \frac{f^{(2021)}(0)}{2021!}, \therefore f^{(2021)}(0) = 2020 \cdot 2020!.$

14. 将一个正数 10 分成三个正数 x, y, z , 使得 $f(x, y, z) = e^x y^2 z^3$ 取值最大.

解: 作拉氏函数 $F = x + 2 \ln y + 3 \ln z + \lambda(x + y + z - 10)$,

$$\text{令} \begin{cases} F_x = 1 + \lambda = 0, \\ F_y = \frac{2}{y} + \lambda = 0, \\ F_z = \frac{3}{z} + \lambda = 0, \\ x + y + z - 10 = 0 \end{cases}$$

得唯一驻点 $x = 5, y = 2, z = 3$; 最大值 $f = 108e^5$.

五 证明题 (10 分)

15. 设 $s(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} x^n$. (1735 年欧拉公布了一个结果: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.)

(1) 求幂级数的收敛域; (2) 求 $s'(x)$;

(3) 证明 $\int_0^1 s(y) dy = \frac{\pi^2}{12} - 2 \ln 2 + 1$.

解: (1) $\rho = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = 1, R = 1$;

$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ 和 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 绝对收敛, 所以收敛域为 $[-1, 1]$.

$$\begin{aligned} (2) s'(x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^{n-1} = -\frac{1}{x} \sum_{n=1}^{\infty} \frac{(-x)^n}{n} = \frac{1}{x} \sum_{n=1}^{\infty} \int_0^x (-x)^{n-1} dx \\ &= \frac{1}{x} \int_0^x \sum_{n=1}^{\infty} (-x)^{n-1} dx = \frac{1}{x} \int_0^x \frac{1}{1+x} dx = \frac{\ln(1+x)}{x}, \quad x \in (-1, 1). \end{aligned}$$

$$(3) \because s(y) = s(0) + \int_0^y s'(x) dx = \int_0^y \frac{\ln(1+x)}{x} dx,$$

$$\begin{aligned} \therefore \int_0^1 s(y) dy &= \int_0^1 \int_0^y \frac{\ln(1+x)}{x} dx dy = \int_0^1 \frac{\ln(1+x)}{x} dx \int_x^1 dy \\ &= \int_0^1 \frac{\ln(1+x)}{x} (1-x) dx = \int_0^1 \frac{\ln(1+x)}{x} dx - \int_0^1 \ln(1+x) dx, \end{aligned}$$

$$\text{其中, } \int_0^1 \frac{\ln(1+x)}{x} dx = s(1) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - 2 \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{\pi^2}{12},$$

$$\int_0^1 \ln(1+x) dx = x \ln(1+x) \Big|_0^1 - \int_0^1 \frac{x}{1+x} dx = 2 \ln 2 - 1.$$

所以有 $\int_0^1 s(y) dy = \frac{\pi^2}{12} - 2 \ln 2 + 1$. 证毕